EQUATOR NETWORK COMPLIANCE DOCUMENT

TRIPOD+AI Reporting Checklist

Transparent Reporting of a Multivariable Prediction Model of Individual Prognosis Or Diagnosis — Artificial Intelligence
Reference: *Collins GS, Moons KGM, Dhiman P, Fewtrell PJ, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use artificial intelligence.* ***BMJ 2024;385:e078378.*** *(equator-network.org)*

Course: BCSE335L Healthcare Data Analytics (Final Review)
Project: DERMA-GUARD (Multimodal Dermatology AI)
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Repository: https://github.com/KavyaPatel83/DERMA-GUARD
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Dataset: PAD-UFES-20 (N=2,298, 1,373 patients, CAAE: 54611416.4.0000.5060)
Compliance Status: 100% Fully Compliant across all 27 checklist items

	TRIPOD+AI Topic	Section & Page	Implementation Details in DERMA-GUARD
TITLE & ABSTRACT			
1	Title	Title & Page 1	Title identifies clinical problem (skin lesion triage), dataset (smartphone dermatology), and AI approach (Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral).
2	Abstract	Abstract (p. 1)	249-word structured summary: Background, Objective, Methods (7-model ladder, QRUC evidence profiler), Results with 95% CIs (89.76% Acc, 0.9985 AUROC, 100% melanoma recall, 3.1% LTT risk), Conclusion.
INTRODUCTION			
3a	Background	Section I (p. 1)	Skin cancer epidemiology (BCC, SCC, melanoma mortality >70% if delayed), diagnostic disparities in primary healthcare, smartphone photography accessibility, and failure modes of uncalibrated deep learning.
3b	Objectives	Section I (p. 1)	Formulates explicit research questions: RQ1 (evidence-aware dynamic weighting), RQ2 (distribution-free LTT risk bounds ≤5%), RQ3 (multimodal visual RAG case grounding & DPDP Act compliance).
3c	Novelty	Section II (p. 1)	5 foundational novelties: (1) Dynamic QRUC Evidence Profiler, (2) Certified LTT Risk Bounds (≤3.1%), (3) 95% Mondrian Conformal Sets, (4) Dual-modality FAISS RAG, (5) Active VOI acquisition policy.
METHODS & MATERIALS			
4a	Data Source	Section IV-A (p. 2)	Retrospective PAD-UFES-20 dataset collected at UFES, Brazil (2017–2019). Distributed under CC BY 4.0; Ethics Committee approval CAAE: 54611416.4.0000.5060.
4b	Eligibility Criteria	Section IV-A & Fig. 1 (p. 2)	Inclusion: smartphone lesion photo with 26 clinical covariates and verified diagnostic outcome. Exclusion: corrupted images, missing ground truth. Total analyzed: N=2,298 lesions across 1,373 unique patients.
5a	Outcome Definition	Section IV-A & IV-E (p. 2)	Primary: 6-class diagnosis (BCC, ACK, NEV, SEK, SCC, MEL). Secondary: binary oncological triage (malignant {BCC, SCC, MEL} vs benign/pre-malignant {ACK, NEV, SEK}).
5b	Ground Truth	Section IV-A & Table II (p. 2)	>58% verified through gold-standard histopathological biopsy; remainder confirmed by unanimous consensus of a panel of three senior board-certified dermatologists.
6a	Candidate Predictors	Section IV-A, IV-C (p. 2)	Visual: 224x224x3 RGB smartphone photography. Metadata: 26 clinical features (age, sex, site, Fitzpatrick type, diameter, bleeding, itching, pain, growth). Text: symptom narrative.
6b	Predictor Handling	Section IV-C (p. 2)	Continuous features scaled with RobustScaler; categorical features one-hot encoded; binary symptoms mapped to {0,1}. Imputations fitted strictly on training fold.
7	Sample Size	Table II (p. 2)	Total N=2,298 lesions across 1,373 unique patients (Train: 1,953; Val: 229; Test: 459). 1,089 malignant events (47.4%). Ratio exceeds 50 events per candidate predictor group.
8a	Missing Data	Section IV-C & IV-E (p. 2)	Missingness documented per feature; median imputed using training statistics. Feature completeness directly modulates Evidence Sufficiency score (S).
8b	Data Leakage Guard	Section IV-B & Fig. 1 (p. 2)	Strict patient-disjoint split by patient_id. Preprocessing transforms, GrabCut bounds, and temperatures fitted exclusively on training set. Zero cross-patient leakage.

	TRIPOD+AI Topic	Section & Page	Implementation Details in DERMA-GUARD
9a	Model Architectures	Section IV-C, IV-D (p. 2)	7 systematic models (A–G): ResNet50 + ViT Self-Attention (Image); 5-layer Residual MetaBlock MLP (Metadata); Bio_ClinicalBERT (Text); AdamW + cosine annealing.
9b	Evidence Fusion	Section IV-E (p. 2)	Formulates Quality (Q_m), Reliability (R_m via TTA entropy), Utility (U_m), Consistency (C), and Sufficiency (S). Dynamic log-space weight update: w_m ← (w_m^0.6 π_m^0.4) / Z.
10a	Calibration	Section IV-F (p. 2)	Post-hoc Temperature Scaling on validation split minimizing NLL; Expected Calibration Error (ECE) evaluated across 10 bins (ECE = 0.040 for Model G).
10b	Conformal Prediction	Section II-C, IV-F (p. 1, 2)	95% Mondrian Conformal Prediction sets C(x) derived via class-conditional non-conformity quantiles, guaranteeing Pr(y ∈ C(x)) ≥ 0.95 under exchangeability.
10c	Risk-Controlled Policy	Section II-B, IV-E, VI (p. 1, 2, 7)	4-tier decision gate (ACCEPT, ARBITRATE, ACQUIRE, ESCALATE) with Learn-then-Test (LTT) risk bound certifying ≤3.1% missed cancer rate (p=0.011 vs. L2D at 4.9%).
RESULTS			
11	Performance Metrics	Section V-A & Table III (p. 3)	Reports AUROC, AUPRC, Sensitivity, Specificity, Macro F1, Accuracy, and ECE — all with Wilson score and 10,000-iteration bootstrap 95% confidence intervals.
12a	Discrimination	Table III, V, VI (p. 3, 4)	Model G (Tri-Modal): 89.76% Acc [95% CI: 86.6, 92.2], Sens 0.9456 [95% CI: 0.918, 0.968], Spec 0.9791, AUROC 0.9985, AUPRC 0.9958. Model D (Image+Meta): 88.45% Acc, 0.9994 AUROC.
12b	Subgroup & Class	Section V-B, V-E (p. 3, 4)	100% recall on Melanoma (10/10), Nevi (49/49), SEK (47/47). Fitzpatrick subgroups: Types I–II (91.2%), Types III–IV (88.4%), Types V–VI (82.6%).
13	Explainability	Section V-E & Fig. 5 (p. 4)	SHAP feature attributions on metadata: Age (0.38), Bleeding (0.34), Lesion Diameter (0.29), Growth (0.26). Grad-CAM visual heatmaps highlight lesion margins.
14	Visual Case RAG	Section IV-F, VI & Fig. 5 (p. 4, 7)	Dual-modality FAISS IVF64 index over 1,953 training cases retrieves top-8 confirmed look-alikes in <10 ms with concordance weighting (91.3% test concordance).
DISCUSSION & DECLARATIONS			
15	Clinical Meaning	Section VI (p. 7)	Interprets multimodal synergy, 4-tier decision triage, analogical proof via RAG look-alikes, and Agentic AI coordination fail-safes against silent diagnostic corruption.
16	Limitations	Section VI-E (p. 7)	Modest test melanoma count (n=10) matching Brazilian prevalence, synthetic text generated from standardized records, Fitzpatrick representation skew (Types V–VI at 5%).
17	Ethics & Regulations	Declarations (p. 8)	UFES Ethics CAAE: 54611416.4.0000.5060; full compliance with Indian DPDP Act 2023 (Sections 4, 6, 8) and ICMR Ethical Guidelines for AI in Healthcare (2023).
18	Code & Data Access	Declarations (p. 8)	Mendeley DOI: 10.17632/zr7vgbcyr2.1. Complete code, weights policy, and 230 automated unit/integration tests hosted at https://github.com/KavyaPatel83/DERMA-GUARD.

Attestation of Compliance: We hereby certify that the manuscript "*DERMA-GUARD: Evidence-Aware Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral for Multimodal Dermatology*" strictly adheres to the **TRIPOD+AI Statement (BMJ 2024;385:e078378)** as catalogued by the EQUATOR Network. All items have been implemented, evaluated, and documented truthfully.
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